

1. MBSF

FILE	Summary	Criteria	File size
MBSF_YR [2012- 2016]	Observations in MBSF Summary File all years	Count of Bene who died each yr 2012- 36,602 2013- 35,348 2014- 34,636 2015- 33,785 2016- 33,093	2012- 999,999 2013- 963,397 2014- 928,049 2015- 893,413 2016- 859,628
MBSF_AGE_YR		Calculate Age using DOB Check against AGE_AT_END_REF_YR Delete if 65 > Age > 75 in 2012 Delete if 66 > Age > 76 in 2013 Delete if 67 > Age > 77 in 2014 Delete if 68 > Age > 78 in 2015 Delete if 69 > Age > 79 in 2016	2012- 999,995 2013- 963,393 2014- 928,045 2015- 893,409 2016- 859,624
MBSF_noADRD_YR		Delete if ADRD Dx = 1 where BENE_ENROLLMT_REF_YR = 2012	2012- 2013- 2014- 2015- 2016-

1.1.1. ADRD Bynum– MEDPar Claims

FILE	Summary	Criteria	File size
MEDPAR_YR [2012- 2016]	Import MedPAR Claims		2012- 381,184 2013- 355,497 2014- 337,601 2015- 332,028 2016- 322,530
ADRD_MEDPAR_YR [2012- 2016]	Estimate ADRD Dx using ICD-9 & ICD-10 Codes	if ICD-9 = ('797' '2900' '2902' '2903' '2904' '2908' '2909' '2940' '2941' '2942' '3310' '3312' '3317' '29010' '29011' '29012' '29013' '29020' '29021' '29040' '29041' '29042' '29043' '29410' '29411' '29420' '29421' '33111' '33119' '33182' '33183')	2012- 39,119 2013- 41,119 2014- 42,864 2015- 45,696 2016- 48,183

		or ICD-10 = ('F0150' 'F0151' 'F0280' 'F0281' 'F0390' 'F0391' 'F04' 'F067' 'G300' 'G301' 'G308' 'G309' 'G310' 'G311' 'G312' 'G3101' 'G3109' 'G3183' 'G3184' 'R4181')	
DIAG_ADRD_MEDPAR_YR [2012- 2016]	Remove duplicate rows based on all columns (Bene_ID, ServiceDt, ServiceThruDt)		2012- 37,388 2013- 39,457 2014- 41,221 2015- 43,958 2016- 46,352
SERVDT_ADRD_MEDPAR_YR [2012- 2016]	Collapse to Service Date Level	Group by bene_id, ServiceDt, ServiceThruDt, ADRD_Dx	2012- 29,067 2013- 30,985 2014- 32,447 2015- 34,892 2016- 37,105
BENE_ADRD_MEDPAR_YR [2012- 2016]	Collapse to Beneficiary Level	Group by bene_id, ADRD_Dx	2012- 16,233 2013- 17,223 2014- 18,031 2015- 18,938 2016- 19,799

1.1.2. ADRD Bynum– Hospice Claims

FILE	Summary	Criteria	File size
HOSPICE_YR [2012- 2016]	Keep Final Bills Only	where clm_query_code ="3" (final bills only)	2012- 16,869 2013- 16,575 2014- 16,077 2015- 15,884 2016- 15,087
ADRD_HOSPICE_YR [2012- 2016]	Estimate ADRD Dx using ICD-9 & ICD-10 Codes	if ICD-9 = ('797' '2900' '2902' '2903' '2904' '2908' '2909' '2940' '2941' '2942' '3310' '3312' '3317' '29010' '29011' '29012' '29013' '29020' '29021' '29040' '29041' '29042' '29043' '29410' '29411' '29420' '29421' '33111' '33119' '33182' '33183') or ICD-10 = ('F0150' 'F0151' 'F0280' 'F0281' 'F0390' 'F0391' 'F04' 'F067' 'G300' 'G301' 'G308' 'G309' 'G310' 'G311' 'G312' 'G3101' 'G3109' 'G3183' 'G3184' 'R4181')	2012- 1,367 2013- 1,794 2014- 2,324 2015- 2,934 2016- 4,161
DIAG_ADRD_HOSPICE_YR [2012- 2016]	Remove duplicate rows based on all columns (Bene_ID, ServiceDt, ServiceThruDt)		2012- 1,366 2013- 1,794 2014- 2,324

			2015- 2,934 2016- 4,161
SERVDT_ADRD_HOSPICE_YR [2012- 2016]	Collapse to Service Date Level	Group by bene_id, ServiceDt, ServiceThruDt, ADRD_Dx	2012- 1,285 2013- 1,669 2014- 2,043 2015- 2,332 2016- 3,147
BENE_ADRD_HOSPICE_YR [2012- 2016]	Collapse to Beneficiary Level Get Min(ServiceDt)	Group by bene_id, ADRD_Dx	2012- 1,229 2013- 1,567 2014- 1,943 2015- 2,237 2016- 3,006

1.1.3. ADRD Bynum– Carrier (FFS) Claims

FILE	Summary	Criteria	File size
CARRIER_YR [2012- 2016]	Combine CARR CLAIMS_YR + LINE_YR	Delete if line_prcsg_ind_cd = ("I", "M", "R") or line_alowd_chrg_amt ≤ 0	2012- 48,490,069 2013- 47,149,317 2014- 45,752,982 2015- 44,920,769 2016- 43,563,798
ADRD_CARRIER_YR [2012- 2016]	Estimate ADRD Dx using ICD-9 & ICD-10 Codes	if ICD-9 = ('797' '2900' '2902' '2903' '2904' '2908' '2909' '2940' '2941' '2942' '3310' '3312' '3317' '29010' '29011' '29012' '29013' '29020' '29021' '29040' '29041' '29042' '29043' '29410' '29411' '29420' '29421' '33111' '33119' '33182' '33183') or ICD-10 = ('F0150' 'F0151' 'F0280' 'F0281' 'F0390' 'F0391' 'F04' 'F067' 'G300' 'G301' 'G308' 'G309' 'G310' 'G311' 'G312' 'G3101' 'G3109' 'G3183' 'G3184' 'R4181')	2012- 399,396 2013- 427,888 2014- 458,156 2015- 514,971 2016- 609,263
DIAG_ADRD_CARRIER_YR [2012- 2016]	Remove duplicate rows based on all columns (Bene_ID, ServiceDt, ServiceThruDt)		2012- 207,160 2013- 226,204 2014- 245,097 2015- 278,787 2016- 340,872
SERVDT_ADRD_CARRIER_YR [2012- 2016]	Collapse to Service Date Level	Group by bene_id, ServiceDt, ServiceThruDt, ADRD_Dx	2012- 170,395 2013- 185,766 2014- 201,806 2015- 228,420 2016- 270,463

1.1.4. ADRD Bynum– Hospital Outpatient (HOF) Claims

FILE	Summary	Criteria	File size
OUTPT_YR [2012- 2016]	Combine HOF CLAIMS_YR + REVENUE_YR		2012- 33,860,119 2013- 32,882,420 2014- 32,307,838 2015- 32,063,939 2016- 31,495,957
HOF_YR [2012- 2016]		RHC_FQHC=0; if &claim_fac_type_code.='7' and &claim_srvc_class_code. in('1', '3') and ('0510' <=&rev_cntr_code.<='0529') then RHC_FQHC=1; CAH_Opt2=0; if (substr(&provider_num.,3,2)='13' and &rev_cntr_code.='0960') then CAH_Opt2=1; */ Delete if RHC_FQHC=0 and CAH_Opt2=0	2012- 1,028,908 2013- 996,383 2014- 974,902 2015- 1,046,701 2016- 1,055,156
ADRD_HOF_YR [2012- 2016]	Estimate ADRD Dx using ICD-9 & ICD-10 Codes	if ICD-9 = ('797' '2900' '2902' '2903' '2904' '2908' '2909' '2940' '2941' '2942' '3310' '3312' '3317' '29010' '29011' '29012' '29013' '29020' '29021' '29040' '29041' '29042' '29043' '29410' '29411' '29420' '29421' '33111' '33119' '33182' '33183') or ICD-10 = ('F0150' 'F0151' 'F0280' 'F0281' 'F0390' 'F0391' 'F04' 'F067' 'G300' 'G301' 'G308' 'G309' 'G310' 'G311' 'G312' 'G3101' 'G3109' 'G3183' 'G3184' 'R4181')	2012- 24,109 2013- 29,856 2014- 33,020 2015- 35,370 2016- 40,518
DIAG_ADRD_HOF_YR [2012- 2016]	Remove duplicate rows based on all columns (Bene_ID, ServiceDt, ServiceThruDt)		2012- 9,660 2013- 12,273 2014- 13,583 2015- 14,479 2016- 16,431
SERVDT_ADRD_HOF_YR [2012- 2016]	Collapse to Service Date Level	Group by bene_id, ServiceDt, ServiceThruDt, ADRD_Dx	2012- 8,061 2013- 10,019 2014- 10,920 2015- 11,186 2016- 12,001

1.1.5. Merge HOF and Carrier Files to keep Claims 7+ days apart

FILE	Summary	Criteria	File size
HOF_CARRIER_YR [2012- 2016]	Append SERVDT_CARRIER_YR + SERVDT_HOF_YR		2012- 178,456 2013- 195,785 2014- 212,726 2015- 239,606 2016- 282,464
INT_HOF_CARRIER_YR [2012- 2016]	Collapse to Beneficiary Level Create – min(ServiceDt) , max(ServiceDt) , time_between_dx= max(ServiceDt) - min(ServiceDt), if max(ServiceDt) - min(ServiceDt) >= 7 then ge7= 1		2012- 38,379 2013- 40,794 2014- 43,016 2015- 47,983 2016- 52,649
BENE_ADRD_HOF_CARRIER_YR [2012- 2016]	No duplicates BENE_IDs	Where ge7= 1 ; (Days btw min and max Dx >=7 days)	2012- 23,711 2013- 26,010 2014- 28,128 2015- 31,503 2016- 35,894

1.1.6. Merge All Claim Type Files

FILE	Summary	Criteria	File size
BENE_ADRD_ALL_SOURCES_YR [2012- 2016]	Carrier HOF, MedPAR, Hospice	ADRD_Dt = min(min_ServiceDt_hof_carrier, min_ServiceDt_medpar, min_ServiceDt_hospice)	2012- 31,846 2013- 34,496 2014- 36,880 2015- 40,148 2016- 44,379
BENE_ADRD_ALL_YR	Check for duplicate BENE_IDs No Duplicates – All rows kept		Same as above

1.1.7. ADRD using GNN (Generic Drug Name) from Pt D Event File

FILE	Summary	Criteria	File size
PDE_YR [2012- 2016]	Import file Rename ADRD_Dx = ADRD_Dx_ICD ADRD_Dt = min_ADRD_Dt		2012- 34,746,954 2013- 34,705,594 2014- 33,163,977 2015- 31,524,675 2016- 30,142,076
ADRD_PDE_YR [2012- 2016]	Create Flag for ADRD Dx based on ADRD drug prescription Keep min date of prescription as proxy for ADRD Dt	ADRD_Dx_PDE =1 if GNN = Memantine Donepezil Rivastigmine Galantamine TACRINE	Same as above
BENE_ADRD_PDE_YR [2012- 2016]	Collapse to Beneficiary Level	Group by Bene_ID, Record_ID	2012- 944,713 2013- 915,636 2014- 885,634 2015- 855,706

			2016- 825,253
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2.1.8 Merge ADRD from Pt D and ICD-9,10 Codes (Bynum)

FILE	Summary	Criteria	File size
BENE_ADRD_ICD_YR [2012- 2016]	Import ADRD Dx derived using ICD-9,10 Codes from Carrier HOF, MedPAR, Hospice (Bynum)		2012- 31,846 2013- 34,496 2014- 36,880 2015- 40,148 2016- 44,379
BENE_ADRD_ICD_PDE_YR [2012- 2016]	Beneficiary Level ADRD Dx based on ADRD drug prescription and ICD-9,10 Codes	If adrd_dx_pde = 1 or adrd_dx_icd = 1 then ADRD_Dx_ICD_PDE = 1;	2012- 945,258 2013- 916,231 2014- 886,232 2015- 856,371 2016- 825,930

3. Tricyclics and Anti-Inflammatory Drugs from Pt D Event File

FILE	Summary	Criteria	File size
PDE_YR [2012- 2016]	Import file and create NDC_Package_Code from PROD_SRVC_ID		2012- 34,746,954 2013- 34,705,594 2014- 33,163,977 2015- 31,524,675 2016- 30,142,076

Non-Selective NSAIDs (17 drugs)

If GNN = Diclofenac, Diflunisal, Etodolac, Fenoprofen, Flurbiprofen, Ibuprofen, Indomethacin, Ketoprofen, Ketorolac, Mefenamic Acid, Meloxicam, Nabumetone, Naproxen, Oxaprozin, Piroxicam, Sulindac, Tolmetin then NON_SELEC_NSAID = 1

Cox-2 Selective NSAIDs

If GNN = Celecoxib then COX2_SELEC_NSAID = 1

Corticosteroids (28 drugs)

If GNN = Budesonide, Beclomethasone, Betamethasone, Celestone, Cutivate, Decadron, Deltasone, Depo-Medrol, Dexamethasone, Diprolene, Diprosone, Entocort Ec, Flonase, Flovent, Fludrocortisone, Kenalog, Medrol, Megace, Megestrol Acetate, Methylprednisolone, Orasone, Prednisolone, Prednisone, Rhinocort, Solu-Medrol, Solumedrol, Triamcinolone Acetonide, Pulmicort then CORTICOSTEROID = 1

Disease Modifying Anti-Rheumatic Drug (DMARD) (19 drugs)

If GNN = Sulfasalazine, Mesalazine, Leflunomide, Deucravacitinib, Apremilast, Abatacept, Bimekizumab, Ixekizumab, Rituximab, Tocilizumab, Sarilumab, Anakinra, Ustekinumab, Secukinumab, Belimumab, Guselkumab, Rizankizumab, Tildrakizumab, Dupilumab then DMARD = 1

Targeted DMARD

If GNN = tofacitinib, abrocitinib, baricitinib, filgotinib, upadacitinib then TARGETED_DMARD = 1

Other Drugs (4 drugs)

If GNN = ciclosporin, cyclosporin, azathioprine, tacrolimus, sirolimus, methotrexate, hydroxychloroquine then OTHER_AI_DRUG = 1

Interferons (IFN) drugs (3 drugs)

If GNN = peginterferon, interferon beta, anifrolumab then IFN_DRUG = 1

Anti-TNF drugs (7 drugs)

If GNN = adalimumab, certolizumab, etanercept, golimumab, infliximab, anti integrin, vedolizumab then ANTI_TNF = 1

if GNN = chlorpromazine then CPZ = 1

if GNN = prochlorperazine then PCPZ = 1

if GNN = trifluoperazine then TRIHYD = 1

if GNN = amitriptyline then AMIHYD = 1

if GNN = perphenazine then PFZ = 1

if GNN = fluphenazine then FLUP = 1

if GNN = promethazine then PROHYD = 1

if GNN = nortriptyline then NORHYD = 1

PDE_DRUGS_YR [2012- 2016]	Identify and group Anti-Inflammatory drugs based on mechanism and Tricyclic drugs	Abovementioned	Same as above
BENE_PDE_DRUGS_YR [2012- 2016]	Collapse to Beneficiary Level	Group by Bene_ID Record_ID	2012- 944,713 2013- 915,636 2014- 885,634 2015- 855,706 2016- 825,253
BENE_PDE_DRUGS1_YR [2012- 2016] <i>Tricyc_ai_drugs_2012</i>	Add Drug Count		Same as above

4.1. Prepare ICD Codes using All Claim Files

FILE	Summary	Criteria	File size
MEDPAR_YR HOSPICE_YR CARR_BASE_YR HOF_BASE_YR [2012- 2016]	Import 4 Claim Types X 5 years		2012- 2013- 2014- 2015- 2016-
COMORB_PREP_YR	Append all 4 Claim Types for each yr. Keep ICD-9/10 Codes	Set MEDPAR_YR HOSPICE_YR CARR_BASE_YR HOF_BASE_YR	2012- 32,989,765 2013- 32,256,893 2014- 31,501,641 2015- 31,069,556 2016- 30,445,230

4.2. Comorbidities from Combined All Claim Type Files

FILE	Summary	Criteria	File size
COMORB_PREP_YR [2012- 2016]	Consolidated Claims w ICD-9/10 codes from MEDPar, Hospice, HOF and Carrier Claims		2012- 32,989,765 2013- 32,256,893 2014- 31,501,641 2015- 31,069,556 2016- 30,445,230
Hypertension if dxcodes(i) in ("4010" "4011" "4019" "40501" "40509" "40511" "40519" "40591" "40599") or dxcodes(i) in: ("I10" "I15") then hypertension = 1 Diabetes			


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if dxcodes(i) in ("24900" "24910" "24920" "24930" "24940" "24950" "24960" "24970" "24980" "24990" "25000" "25001"
"25010" "25011" "25020" "25030" "25031" "25040" "25041" "25050" "25051" "25060" "25061" "25070" "25071" "25080"
"25081" "25090" "25091" "3572" "36201" "36202" "36204" "36205" "36206" "36207" "36641") or
dxcodes(i) in: ("E10" "E11" "E13") then diabetes = 1

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Stroke

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if dxcodes(i) in ("43301", "43311", "43321", "43331", "43381", "43391", "43401", "43411", "43491") or
dxcodes(i) in: ("I63") then stroke = 1

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Heart Disease

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if dxcodes(i) in ("40200" "40201" "40210" "40211" "40290" "40291" "4111" "412" "4139" "41401" "41402" "41403" "41404"
"41405" "41406" "41407" "41410" "41411" "41412" "41419" "4142" "4143" "4144" "4148" "4149" "4292") or
dxcodes(i) in: ("I25" "I11") then heart = 1

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Depression

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if dxcodes(i) in ("29620" "29621" "29622" "29623" "29624" "29625" "29626" "29630" "29631" "29632" "29633" "29634"
"29635" "29636" "29682" "29699" "2980" "311") or
dxcodes(i) in: ("F32" "F33") then depression = 1

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Traumatic Brain Injury (TBI)

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if dxcodes(i) in ("3485" "73382" "80000" "80050" "80100" "80150" "8020" "8021" "80220" "80221" "80222" "80223"
"80224" "80225" "80226" "80227" "80228" "80230" "80231" "80232" "80233" "80234" "80235" "80236" "80237" "80238"
"8024" "8025" "8026" "8027" "8028" "8029" "80300" "80350" "8500" "85011" "85012" "8502" "8503" "8504" "8505"
"85141" "85142" "85143" "85144" "85145" "85146" "85161" "85162" "85163" "85164" "85165" "85166" "85181" "85182"
"85183" "85184" "85185" "85186" "85201" "85202" "85203" "85204" "85205" "85206" "85221" "85222" "85223" "85224"
"85225" "85226" "85241" "85242" "85243" "85244" "85245" "85246" "85301" "85302" "85303" "85304" "85305" "85306"
"85401" "85402" "85403" "85404" "85405" "85406" "87363" "9050" "9064" "9070" "9251" "V1552" "V5419" "V5889"
"V8001") or
dxcodes(i) in: ("S02" "S06" "S07" "Z13850" "Z87820") then tbi = 1

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Smoking

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if dxcodes(i) in ("2920" "29289" "2929" "3051" "V1584" "V1586" "V1589" "V690" "V691" "V692" "V693" "V694" "V698"
"V699" "V7101" "V7102" "V8701" "V8702" "V8709" "V8711" "V8712" "V8719" "V872" "V8731" "V8732" "V8739") or
dxcodes(i) in: ("F17" "Z72" "Z77") then smoking = 1

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Psychosis

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if dxcodes(i) in ("29510" "29520" "29530" "29540" "29560" "29570" "29580" "29590" "29600" "29601" "29602" "29603"
"29604" "29605" "29606" "29681" "2970" "2971" "2972" "2973" "2983" "2984" "2988" "2989" "30122") or
dxcodes(i) in: ("F20" "F21" "F22" "F23" "F24" "F25" "F26" "F27" "F28" "F29" "F30") then psychosis = 1

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Rheumatoid Arthritis

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if dxcodes(i) in ("71481", "7142", "3596", "7140", "3571", "7141", "71489", "7149") or
dxcodes(i) in: ("M05" "M06" "M061") then rheum_arthritis = 1

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COMORB_YR [2012- 2016]	Identify Comorbidities of interest from consolidated claims data w all 4 Claim Types for each yr.		2012- 32,989,765 2013- 32,256,893 2014- 31,501,641 2015- 31,069,556 2016- 30,445,230
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BENE_COMORB_YR [2012- 2016]	Collapse to Beneficiary Level	Group by Bene_ID, Record_ID	2012- 975,805 2013- 943,006 2014- 910,175 2015- 877,473 2016- 845,282
BENE_COMORB1_YR [2012- 2016] <i>bene_comorb_yr</i>	Create flag for any comorbidity and count		2012- 975,805 2013- 943,006 2014- 910,175 2015- 877,473 2016- 845,282

5. Merge All Files – MBSF, ADRD Dx, Comorbidities, Prescription Drugs

FILE	Summary	Criteria	File size
MEDPAR_YR HOSPICE_YR CARR_BASE_YR HOF_BASE_YR [2012- 2016]			2012- 2013- 2014- 2015- 2016-
COMORB_YR	No duplicate BENE_IDs All rows kept		2012- 2013- 2014- 2015- 2016-